BhooDrishti - Technical Design Document & System Architecture

Architectural Overview, 7-Stage Pipeline, Mathematical Models & Database Schema

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1. Executive Summary

In Himachal Pradesh, 86.07% of operational holdings are small or marginal (<2 ha), and pests or diseases threaten 30-40% of yield before symptoms are obvious. BhooDrishti uses a three-tier architecture implementing the core loop: DETECT (Satellite NDVI/NDRE) -> VALIDATE (IoT + Leaf Doctor) -> ACT (Advisory + Mandi Rate Calculator).

2. System Architecture Overview (3-Tier Table)

Architectural Layer	Core Components	Data Flow & Output
1. Data Ingestion Layer	Copernicus Sentinel-2 L2A API, ESP32 + LoRa Sensors (Soil Moisture, Temp, Humidity), Weather APIs, eNAM/Agmarknet	Ingests multispectral bands, real-time ground telemetry, weather forecasts, and mandi prices.
2. Processing, AI & Storage Layer	FastAPI (Python) & Node.js, Scikit-Learn (K-Means), MobileNetV3 CNN (PlantVillage), LLM API, PostgreSQL + GeoJSON	Computes NDVI/NDRE zones, fuses satellite + LoRa data, runs leaf disease inference, and stores spatial records.
3. Farmer Application Layer	Remix (React), Tailwind CSS, Leaflet.js Zoned Map, Voice-Guided In-Field Navigator, Mandi Rate Calculator	Displays stress maps, guides farmers to stressed zones via voice compass, and provides multilingual advisories.

3. End-to-End 7-Stage Pipeline & User Journey

- 1**. Satellite Imagery**: Copernicus Sentinel-2 Level-2A (S2 L2A) multispectral data generates wide-coverage NDVI and NDRE indices.
- 2**. Field Sensor Data**: ESP32 + LoRa sensors collect soil moisture, temperature, and humidity and send field data to the gateway.

- 3. **Weather & Agricultural Data:** Integrates rainfall, temperature, humidity forecasts, and soil parameters.
- **4. AI / Data Processing:** Fuses Sentinel-2 NDVI/NDRE + LoRa Data + Weather Data using K-Means clustering to assess overall crop health.
- **5. Crop Health & Stress Map:** Zoned field visualization on Leaflet.js highlighting Healthy, Moderate, and High-Stress/Deficiency zones.
- **6. BhooDrishti Dashboard & In-Field Navigator:** Mobile/web app shows stress alerts in regional languages; voice-guided compass directs the farmer from their live location straight to the flagged stress zone.
- **7. Smart Advisory & Action:** Farmer uploads leaf photos of the affected zone to Leaf Doctor (MobileNetV3 + LLM) for allopathic + desi remedies, and uses the Mandi Rate Calculator to compare market prices and transport costs.

4. PostgreSQL + GeoJSON Database Schema

- **farmers:** id (UUID), full_name, mobile_number,dob,password,kisan_id,state, district,city,tehsil,patwar_circle,village,primary_crop, last_login,updated_at,location, preferred_language, created_at
- **user_lands:** id (UUID), user_id (FK), crop_name, farmer_name, crop,area_acres,area_sqm,health_score,points,center,notes,created_at,updated_at,subscription_tier,irrigation_status,current_moisture,optimal_treshold
- **structure_tracking_logs:** id (UUID), user_id (FK), farm_id(FK)farm_name,satellite_source, baseline_date, target_date, detected_changes_pct, change_type, risk_level, bounding_box, notes, created_at
- **land_reports:** id (UUID), user_id, land_id, farm_name, farmer_name, crop, rain_chance, advisory_summary, irrigation_schedule, fertilizer_recommendation, pest_risk, created_at, soil_moisture,ndre_score,ndwi_score,temperature,humidity, health_score,soil_n,soil_p,soil_k,soil_ph
- **field_reports:**id,user_id,farm_id,crop,state,district,area_acres,area_sqm,temperature, humidity,rain_chance,precipitation,soil_n,soil_p,soil_k,soil_ph,soil_moisture,ndre_score,ndwi_score,health_pct, stress_pct,coordinates,created_at
- **diagnostic_logs:** id,confidence,symptoms,remedy_type,created_at
- **advisories:**id,state,area_acres,summary,irrigation_schedule,nitrogen_recommendation,created_at